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328653(28)

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BE (6th Semester)

Examination, April-May, 2016

(New Scheme)

Microcontroller and Embedded Systems

Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Solve any two parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

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1. (*a*) What is OFF chip and ON chip concept in microcontroller. 2

(*b*) Write a program to toggle the bit of port P1 and Port P2 continuously (*i*) using AAH and 55 H and (*ii*) using CPL instruction. 7

(*c*) Draw and explain Architecture of 8051 microcontroller.. 7

(*d*) What are the different criteria for choosing the microcontroller for a particular application? 7

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2. (*a*) How can we double the band rate by using TIMER registers? 2

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(*b*) Using Timer 1 in mode 1, generate a time delay of 2 msec.. 7

(*c*) Write a program to generate a square wave of 2 kHz frequency on P1.5. XTAL = 11.0592 MHz. 7

(*d*) Explain in detail significance and role of TCON register. 7

3. (*a*) What is SFR? 2

(*b*) Find the TH1 value (in both decimal and hexadecimal) to set the band rate to each of the following :

(*i*) 9600

(*ii*) 4800 if SMOD = 1.

Assume XTAL = 11.0592 MHz. 7

(*c*) Write a Program, using interrupts to do the following :

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(*i*) Receive data serially and send it to PO

(*ii*) Read port P1, transmit the data serially and copy the content to P2 7

(*d*) Explain MAX 232 and its interfacing with 8031 and 8051. 7

4. (*a*) Write the significance of SOC and EOC pin in 8051. 2

(*b*) Draw and explain stepper motor interfacing with 8051. 7

(*c*) List the features of Real Time embedded system. 7

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(*d*) Draw detail interfacing diagram of ADC 0808 with 8051. 7

5. (*a*) Define Embedded system. 2

(*b*) Discuss the Real life Examples of Embedded system. 7

(*c*) What are different development issues in Embedded system? Discuss them with example. 7

(*d*) Write a note on Embedded design challenges. 7

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